Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 05

Subject: BSC102/AP | **Date: 26/10/2026**
Max Marks: 10 | **Deadline: 13/11/2026**

CO5: Learners will be able to RELATE the foundations of quantum mechanics with the development of modern technology.

Problem 1: de Broglie Hypothesis and Wavelength Derivation

1.1 Statement of de Broglie Hypothesis

In 1924, Louis de Broglie proposed that nature exhibits symmetry between radiation (waves) and matter (particles). Since electromagnetic radiation demonstrates both wave-like characteristics (interference, diffraction) and particle-like characteristics (photoelectric effect), moving material particles must also exhibit wave properties under appropriate conditions.

These waves associated with moving material bodies are referred to as **matter waves** or **de Broglie waves**.

1.2 Deduction of the Expression for de Broglie Wavelength

Consider a photon of frequency ν and wavelength λ. According to Planck's quantum hypothesis, the energy of the photon is:

$$E = h\nu = \frac{hc}{\lambda} \tag{1}$$

where h is Planck's constant (6.626×10^{-34} J $\cdot$ s) and c is the speed of light in vacuum.

According to Einstein's mass-energy equivalence relation, if the photon is treated as having an effective mass m:

$$E = mc^2 \tag{2}$$

Equating equation (1) and equation (2):

$$mc^2 = \frac{hc}{\lambda} \implies mc = \frac{h}{\lambda}$$

Since the linear momentum of the photon is $p = mc$, this yields:

$$\lambda = \frac{h}{p} \tag{3}$$

By extension of the matter-wave hypothesis, de Broglie proposed that equation (3) holds universally for any material particle of rest mass m traveling with non-relativistic velocity v and linear momentum $p = mv$:

$$\lambda = \frac{h}{mv} = \frac{h}{p} \tag{4}$$

1.3 Alternate Expression in Terms of Kinetic Energy and Potential

Let a particle of mass m have kinetic energy K. Since $K = \frac{p^2}{2m}$, the momentum is $p = \sqrt{2mK}$. Therefore:

$$\lambda = \frac{h}{\sqrt{2mK}} \tag{5}$$

If a charged particle carrying charge q is accelerated through a potential difference V, the acquired kinetic energy is $K = qV$. Substituting this gives:

$$\lambda = \frac{h}{\sqrt{2mqV}} \tag{6}$$

Problem 2: Wavelength of an Accelerated Electron ($V = 100$ V)

2.1 Formula and Physical Constants

The de Broglie wavelength of an electron accelerated from rest through an electric potential difference V is given by:

$$\lambda = \frac{h}{\sqrt{2m_e eV}}$$

where:

- Planck's constant: $h = 6.626 \times 10^{-34}$ J $\cdot$ s
- Electron mass: $m_e = 9.109 \times 10^{-31}$ kg
- Elementary charge: $e = 1.602 \times 10^{-19}$ C
- Accelerating potential: $V = 100$ V

2.2 Step-by-Step Calculation

First, calculate the denominator $p = \sqrt{2m_e eV}$:

$$\begin{aligned} p &= \sqrt{2 \times (9.109 \times 10^{-31}\ \text{kg}) \times (1.602 \times 10^{-19}\ \text{C}) \times 100\ \text{V}} \\ &= \sqrt{2.9185 \times 10^{-47}} \\ &= 5.4023 \times 10^{-24}\ \text{kg} \cdot \text{m/s} \end{aligned}$$

Now, compute the wavelength:

$$\begin{aligned} \lambda &= \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{5.4023 \times 10^{-24}\ \text{kg} \cdot \text{m/s}} \\ &= 1.2265 \times 10^{-10}\ \text{m} = 0.1227\ \text{nm} = 1.227\ \text{Å} \end{aligned}$$

Using the standard simplified formula for electrons ($\lambda = \frac{1.227}{\sqrt{V}}$ nm):

$$\lambda = \frac{1.227}{\sqrt{100}}\ \text{nm} = \frac{1.227}{10}\ \text{nm} = 0.1227\ \text{nm} \quad (1.227\ \text{Å})$$

Problem 3: Heisenberg's Uncertainty Principle

3.1 Statement

Formulated by Werner Heisenberg in 1927, the Uncertainty Principle states that it is physically impossible to simultaneously determine both the exact position and exact linear momentum of a microscopic particle along the same spatial direction with arbitrary precision.

Mathematically, for position coordinate x and corresponding linear momentum p_x:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi} \tag{7}$$

where:

- Δx is the uncertainty in measuring position,
- Δp_x is the uncertainty in measuring momentum along the x-axis,
- h is Planck's constant, and $\hbar = \frac{h}{2\pi} \approx 1.055 \times 10^{-34}\ \text{J} \cdot \text{s}$.

3.2 Physical Explanation

1. **Wave Packet Nature:** In quantum mechanics, a localized particle is represented by a wave packet formed by the superposition of waves of slightly different wavelengths. To localize the particle within a very small region Δx, a broader spectrum of wave numbers k (and therefore momenta $p = \hbar k$) must be superposed, which inherently increases Δp.

2. **Measurement Intrusion:** Any physical attempt to observe a microscopic object requires probing it with a photon. Shorter probe wavelengths yield higher spatial accuracy ($\Delta x \sim \lambda$), but high-frequency photons possess large momentum ($p = h/\lambda$), transferring an uncontrollable recoil momentum to the particle and causing larger uncertainty Δp.

3. **Energy-Time Conjugate Pair:** An analogous uncertainty relation holds between energy E and time interval Δt:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

Problem 4: Calculation of Uncertainty in Position

4.1 Given Data

- Measured speed of electron: $v = 4 \times 10^5$ m/s
- Measurement accuracy: $\eta = 0.002\% = \frac{0.002}{100} = 2 \times 10^{-5}$
- Mass of electron: $m = 9.109 \times 10^{-31}$ kg
- Planck's constant: $h = 6.626 \times 10^{-34}$ J $\cdot$ s

4.2 Uncertainty in Velocity and Momentum

The uncertainty in the velocity measurement is:

$$\begin{aligned} \Delta v &= v \times \eta \\ &= (4 \times 10^5 \text{ m/s}) \times (2 \times 10^{-5}) \\ &= 8 \text{ m/s} \end{aligned}$$

The corresponding uncertainty in linear momentum is:

$$\begin{aligned} \Delta p &= m\Delta v \\ &= (9.109 \times 10^{-31} \text{ kg}) \times (8 \text{ m/s}) \\ &= 7.2872 \times 10^{-30} \text{ kg} \cdot \text{m/s} \end{aligned}$$

4.3 Calculation of Minimum Uncertainty in Position

From Heisenberg's uncertainty relation $\Delta x \cdot \Delta p \geq \frac{\hbar}{2} = \frac{h}{4\pi}$:

$$\begin{aligned} \Delta x_{\text{min}} &= \frac{h}{4\pi\Delta p} \\ &= \frac{6.626 \times 10^{-34} \text{ J} \cdot \text{s}}{4\pi \times (7.2872 \times 10^{-30} \text{ kg} \cdot \text{m/s})} \\ &= \frac{6.626 \times 10^{-34}}{9.1574 \times 10^{-29}} \\ &\approx 7.236 \times 10^{-6} \text{ m} = 7.24 \ \mu\text{m} \end{aligned}$$

Alternative convention note: Using the relation $\Delta x \cdot \Delta p \geq \frac{h}{2\pi}$:

$$\Delta x = \frac{h}{2\pi \Delta p} \approx 1.45 \times 10^{-5} \text{ m} = 14.5 \ \mu\text{m}$$

Using the fundamental limit $\Delta x \geq \frac{\hbar}{2}$, the minimum uncertainty in position is 7.24 $\mu\mathbf{m}$.